INCH-POUND

MIL-S-3788G
8 February 1991
SUPERSEDING
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23 January 1986

MILITARY SPECIFICATION

SHAPER, WOODWORKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single and double spindle woodworking shapers of the floor-mounted type with equipment and accessories specified herein. Requirements set forth in this specification and the completed ordering data (see paragraph 6.2) establish the Government's minimum requirements for each procurement.

1.2 Classification. The shapers shall be of the following types and sizes. The types and sizes to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Single Spindle

Size 1 - 1 hp - 20-inch by 26-inch table
Size 2 - 2hp - 24-inch by 36-inch table
Size 3 - 5 hp - 35-inch by 42-inch table

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3220

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Type II - Double Spindle

Size 1 - 2 hp (each spindle) - 12-inch, center-to-center, between spindles, 26-inch by 36-inch table
Size 2 - 2 hp (each spindle) - 18-inch, center-to-center, between spindles, 30-inch by 48-inch table
Size 3 - 3 hp (each spindle) - 24-inch, center-to-center, between spindles, 30-inch by 60-inch table
Size 4 - 3 hp (each spindle) - 30-inch, center-to-center, between spindles, 30-inch by 66-inch table

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111- 5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI 01.1a - Safety Requirements for Woodworking Machinery. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA ICS 6 - Enclosures for Industrial Controls and Systems. (000 adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete shaper shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The shaper shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The shaper shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the shapers shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The shapers and their components that directly consume energy in normal operation shall be designed and constructed for

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energy efficiency in accordance with the latest developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the shapers, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI 01.1. If a conflict arises between 29 CFR 1910 and ANSI 01.1, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial shapers supplied to the commercial market shall be provided for the point of operation and all other parts of the shapers that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The shapers shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the shapers.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device and shall be in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective replacement parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts and fit/function with mating components.

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3.3 Construction. The shapers shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the shaper to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent unintentional change of tightness. Those subject to removal or adjust shall not be swaged, peened, welded, staked, or otherwise permanently deformed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding. brazing. or soldering. Melding, brazing, or soldering shall be employed only where specified in the original design. Any excess material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations. None of these operations shall be employed as a repair measure for defective parts.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the shapers and their related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that will impart the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

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3.4 Components. All shapers covered herein shall consist essentially of a base, table, fence, spindles, spindle drive, electrical system and their related controls.

3.4.1 Base. The base shall be constructed of a mass with rigidity to support the table, motor, spindle and spindle drive mechanism. The base shall contain not less than three holes for mounting to the floor or foundation.

3.4.2 Table. The table shall conform to the dimensions specified in table I. The workpiece support surface of the table shall be flat within 0.003 inch per foot. The spindle hole through the table shall receive one or more table rings to facilitate undercutting operations. The table rings shall be so constructed that they are easily removed and installed and conform to the workpiece support surface of the table. The table shall have mounted shaper guards. The shaper table shall be constructed to provide an exhaust opening for dust and chips. An adapter of the size and shape specified (see 6.2), shall connect to the dust collection system.

3.4.2.1 Fence. The shaper shall have a fence that guides the edge of the workpiece as it is hand-fed toward and away from the cutter. The fence shall conform to table I and allow the use of all cutters within the capacity of the shaper. The fence support shall provide adjustments and means of securely locking in adjusted positions.

3.4.3 Spindles. Spindles shall be constructed and balanced to minimize vibration and to insure centering of Installed cutters. Means shall be provided to prevent spindle rotation during cutter installation and removal. Unless otherwise specified (see 6.2), each spindle shall be independently driven by a belt, pulley and motor to provide both spindle speeds and vertical spindle adjustment in accordance with table I.

3.4.3.1 Spindle, type I, size 1. Type I, size 1 shaper spindle shall be constructed to accept cutters of 1/2 inch and 3/4 inch bore. The spindle shall be supported by ball or roller bearings of the lubricated and sealed-for-life type. Two separate spindles, which interchange, shall also be acceptable to accommodate both the 1/2 inch and 3/4 inch cutter bores. A handwheel located convenient to the operator shall provide vertical adjustment of the spindle in accordance with table I. Vertical spindle axis alignment shall be maintained at a 90 degree angle to the table top surface over the entire spindle adjustment range.

3.4.3.2 Spindles, type 1. sizes 2 and 3 and type II shapers. Type I, sizes 2 and 3 and type II shaper spindles shall be constructed with a spindle diameter conforming to table I. The cutter and collar surface of the spindle shall either be an extension of the spindle and adapter or a detachable spindle extension mounted on the spindle. The spindle shall be supported by ball or roller bearings with means of lubrication in accordance with 3.2.9. Type I, sizes 2 and 3 and type 11 shaper spindles shall be supported

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vertically, with means to maintain the vertical axis alignment of the spindle at a right angle to the table top surface.

3.4.3.3 spindle brake. Unless otherwise specified (see 6.2), type I, size 3 and type II shaper spindles shall be provided with a brake. Manual control shall shut-off the motor and stop spindle rotation within a time period of 10 seconds. An interlock shall be provided in the control circuit to prevent operation of the spindle drive motor while the spindle brake is applied.

3.4.3.4 Spindle positioning mechanism. Spindle positioning shall be accomplished by means of a handwheel located convenient to the operator. Means shall be provided to retain the spindle in any adjusted position over the entire vertical adjustment range. When specified (see 6.2), type I, sizes 2 and 3, and type II shapers shall have a spindle position dial graduated in increments not greater than 0.006 inch.

3.4.4 Electrical requirements. The machine shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 230/460-volts, 3-phase, 60 Hertz (Hz). Unless otherwise specified (see 6.2), the shaper shall be initially wired for operation on 460 volts. Electrical enclosures shall conform to NEMA ICS 6.

3.4.4.1 Motors. Motors shall be rated for 60 min. duty cycle and shall have ball *or* roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof, totally enclosed, fan-cooled enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the shapers shall meet the requirements of table 1 for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The shaper shall be capable of performing surface shaping on wood workplaces within cutter capacity including straight, concave or convex cutters mounted on the spindle. Performance of the shaper shall be such that straight, smooth, shaper surfaces corresponding to the cutter profile shall be produced on wood workplaces.

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3.7 Alignment accuracies. Each shaper shall meet the following alignment tests (see 4.7.3):

a. Spindle runout on the cutter mounting surface shall not exceed 0.004 inch TIR.

b. Spindle end play shall not exceed 0.005 inch.

c. The spindle axis shall be square to the table workpiece support surface within 0.010 inch per foot TIR.

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below:

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

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3.12 Workmanship. Workmanship of the shaper and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to-perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with al 1 requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

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4.5 Quality conformance inspection Quality conformance inspection shall be applied to each shaper prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examinations, all tests, and the inspection to be accepted.

4.6 Examination. Each shaper shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit and 20 to 95 percent relative humidity.

4.7.2 Operational test. Each shaper shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment tolerance test Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.7. The alignment tolerance test shall be perfomed prior to and subsequent to the performance test(s) given in 4.7.4. Any machine adjustments required to comply with 3.7 during or after the performance test shall be cause for rejection.

4.7.4 Performance tests. Performance tests shall be in accordance with 3.6. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The shapers covered by this specification are intended for use in woodworking shops where molding and shaping operations are required,

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Describe the size and shape of adapter required (see 3.4.2).

k. Drive and spindle speeds, if different (see 3.4.3).

l. Spindle brake, if different (see 3.4.3.3).

m. Spindle position dial, if required (see 3.4.3.4).

no Electrical requirements, if different (see 3.4.4).

o. Initial voltage, if different (see 3.4.4).

p" Motors, if different (see 3.4.4.1).

q. Size and capacity, if different (see 3.5).

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r. Specify physical size limitation, if required (see 3.5.1).

s. Specify and fully describe optional equipment, if required (see 3.9).

t. Specify applicable repair parts, if required (see 3.10).

u. Lubrication plate, if required (see 3.11.1).

v. Information to be provided on lubrication plate, if different (see 3.11.1).

w. Nameplate, if different (see 3.11.2).

x. Inspection conditions, if different (see 4.3).

y. First article inspection, if different (see 4.4).

z. Quality conformance inspection, if different (see 4.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DO Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
6.3.1	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Of-The-Shelf (COTS) Manuals	Require all data be in the English language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DO Form 1423.

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6.3.1 Technical data It should be specified in the “Remarks” section of the DD Form 1423 that ‘all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor’s current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D’Unites (SI)," of the International Bureau of Heights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

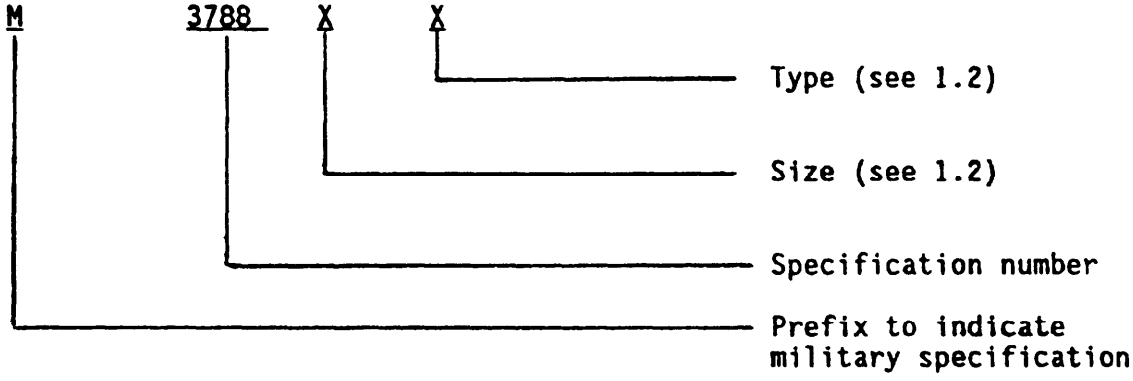
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6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.
Carpentry shop
Double spindle
Floor mounted
Single spindle
Machine

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TABLE I - Sizes and capacities.

Characteristics	Requirement Type I		
	Size 1	Size 2	Size 3
Table:			
Dimension, length by width, minimum inches	20 x 26	24 x 36	35 x 42
Spindle hole dia., inches	6	4	6
Table height, inches	33 to 36	33 to 36	33 to 36
Fence dimension, inches	3 x 12	3 x 12	3 x 12
Spindles:			
Outside, dia., inches	1/2	1	1
Vert. adjustment, inches	3	3	3
RPM, minimum	7000	7000	7000
Motor horsepower:			
Standard	1	2	5
Optional	1-1/2, 2 or 3	3 or 5	7-1/2 or 10

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TABLE I - Sizes and capacities. (continued)

Characteristics	Requirement Type II			
	Size 1	Size 2	Size 3	Size 4
Table:				
Dimension, length by width, minimum inches	26 x 36	30 x 48	30 x 60	30 x 66
Spindle hole dia.,inches	6	6	6	6
Table height, inches	33 to 36	33 to 36	33 to 36	33 x 36
Fence dimension, inches	3 x 12	3 x 12	3 x 12	3 x 12
Spindles:				
Outside, dia., inches	1-1/8	1-1/8	1-1/8	1-1/8
Vert. adjustment, inches	3	3	3	3
RPM	7000	7000	7000	7000
Center to center, inches	12	18	24	30
Motor horsepower:				
Standard	2	2	3	3
Optional	3 or 5	3 or 5	5, 7-1/2 or 10	5, 7-1/2 or 10

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84

User activities:
Army - SM
Navy - YD, MC

Preparing activity:
DLA - IP

Project (3220-0162)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-3788G	910208

3. DOCUMENT TITLE
SHAPER, WOODWORKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. Defense Industrial Plant Equipment Center ATTN: DIPEC-SSM	b. TELEPHONE (Include Area Code) (1) Commercial: 901-775-4797	(2) AUTOVON: AV 683-4797
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:	

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